BREADTH Scholars Curriculum

School of Engineering
BROWN UNIVERSITY

AI Monitoring Through Computer Vision:

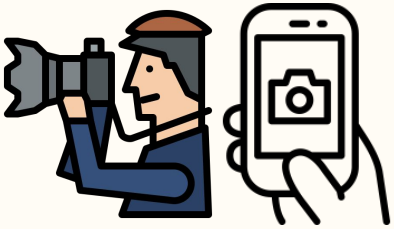
Image Acquisition & Image Processing Basics

Prof. Benjamin Kimia / Chiang-Heng Chien



What is Computer Vision?

Computer vision is an analysis of digital images by a computer for various applications.







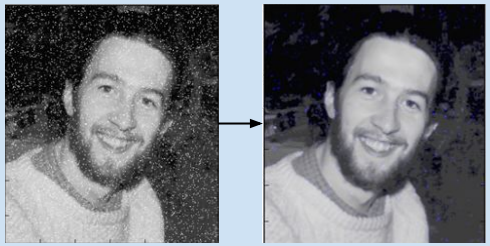






Stages of Computer Vision

Stage 1



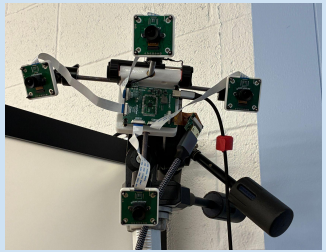
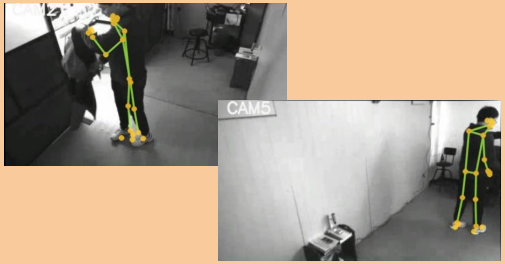
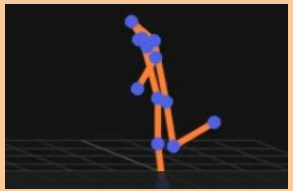


Image processing & hardware

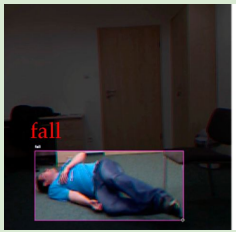
Stage 2



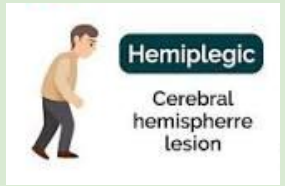


Multi-camera human pose perception & 3D reconstruction

Stage 3





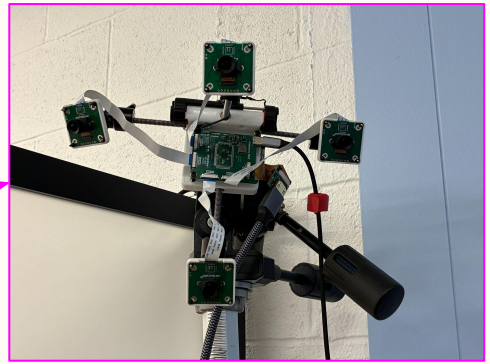


Fall detection & gait analysis

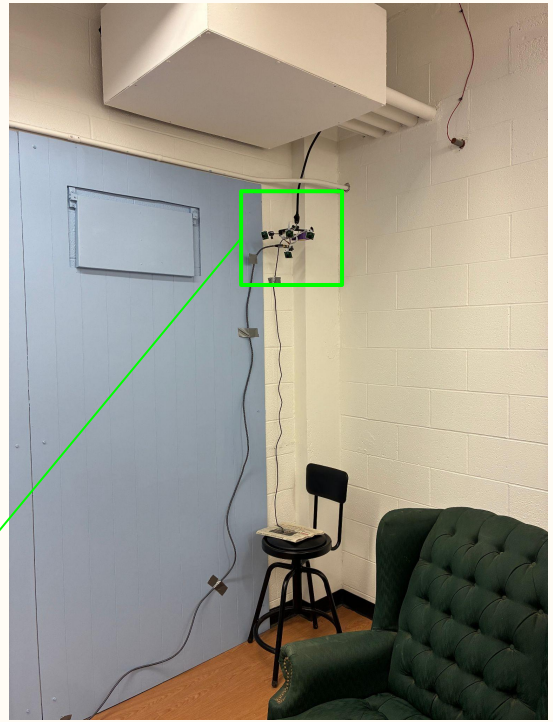
Our Goal: Multiview Gait Analysis for Abnormal Gait Recognition

In the Past: We build our own fall detection dataset in Sandbox









Our Goal: Multiview Gait Analysis for Abnormal Gait Recognition

In the Past: We build our own fall detection dataset in Sandbox

- 25 videos @ 40 FPS.
- Captured by Chiang-Heng, Jieye and Yuan.
- Annotated by Tangtangfang.



Our Goal: Multiview Gait Analysis for Abnormal Gait Recognition

Now: Real-World Multiview Dataset Construction

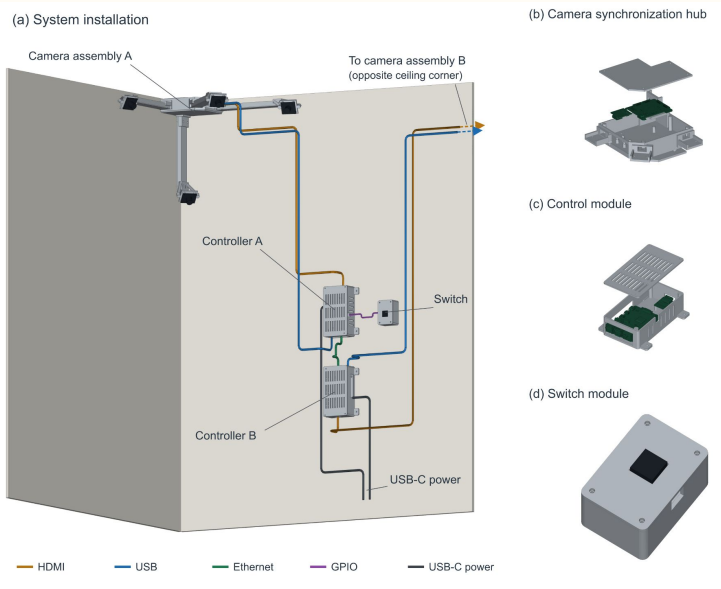
BROWN-BEAR Dataset: Brown **R**eal-World **O**bservation for **W**ellness and **N**euromotor **B**ehavioral **E**valuation and **A**ctivity **R**ecordings







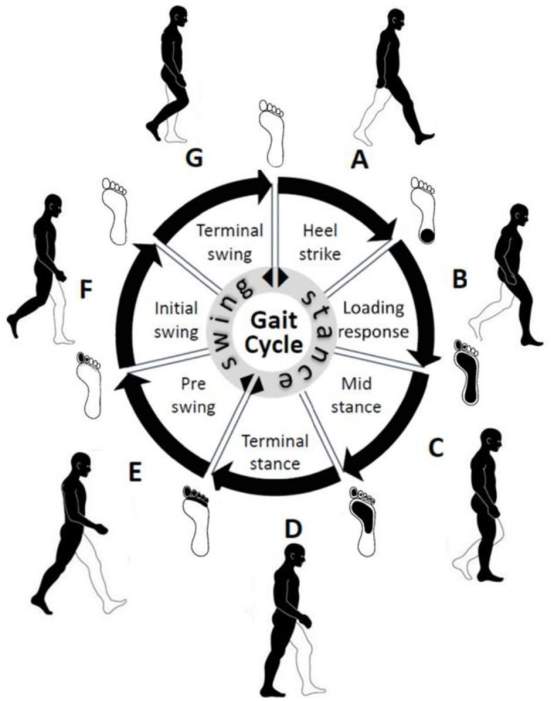




Our Goal: Multiview Gait Analysis for Abnormal Gait Recognition

Multiview Fall Detection & Abnormal Gait Detection/Recognition





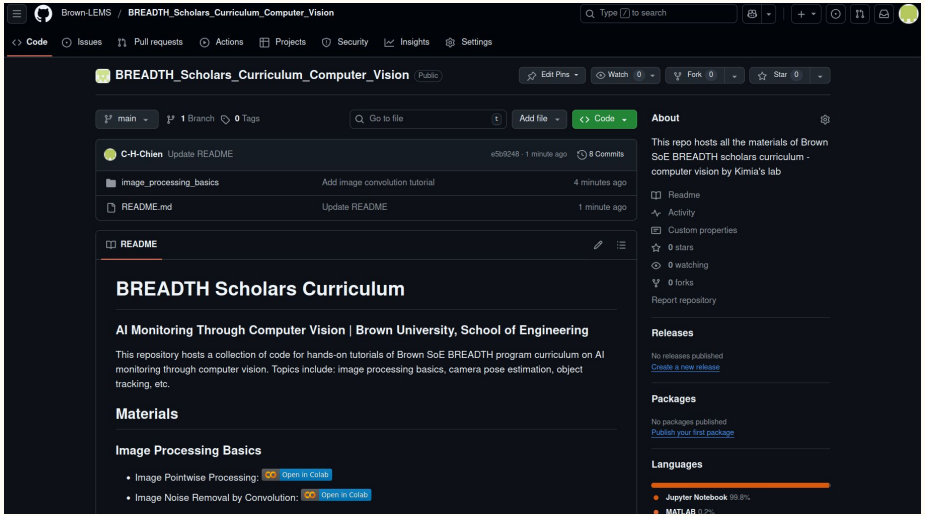
Knowledge + Practical Implementation

PART I: Image Formation & Acquisition

- Practical Implementation: Image Pointwise Processing

Part II: Image Convolution

- Practical Implementation: Image Noise Removal

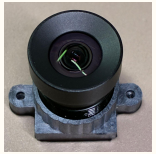
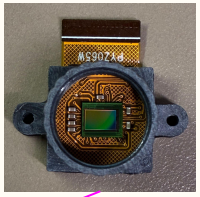


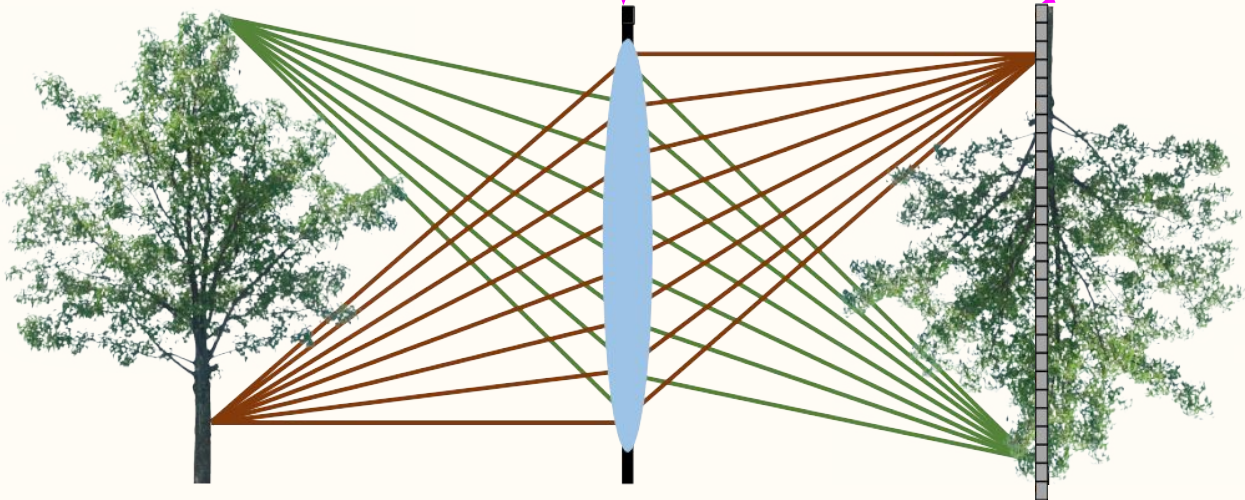
https://github.com/Brown-LEMS/BREADTH_Scholars_Curriculum_Computer_Vision

PART I: Image Formation & Acquisition

What is a (digital) image?

Where do images come from?

Optical camera  = camera lens  + digital sensor 



PART I: Image Formation & Acquisition

Where do images come from?

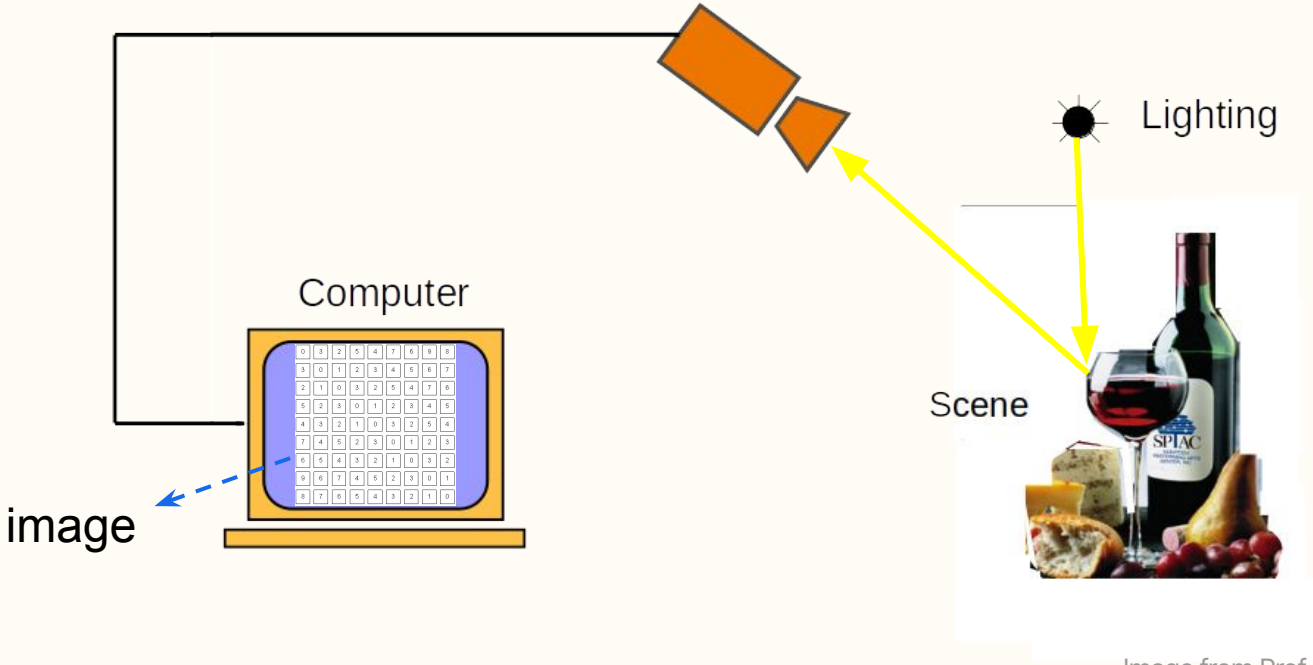


Image from Prof. Srinivasa Narasimhan's slide

PART I: Image Formation & Acquisition

What is an image?

Color (RGB) or Gray
Image “intensity” is encoded by [0, 255]
(Unsigned integer 8-bit)

pixels

0	3	2	5	4	7	6	9	8
3	0	1	2	3	4	5	6	7
2	1	0	3	2	5	4	7	6
5	2	3	0	1	2	3	4	5
4	3	2	1	0	3	2	5	4
7	4	5	2	3	0	1	2	3
6	5	4	3	2	1	0	3	2
9	6	7	4	5	2	3	0	1
8	7	6	5	4	3	2	1	0

image

Why 8-bit?

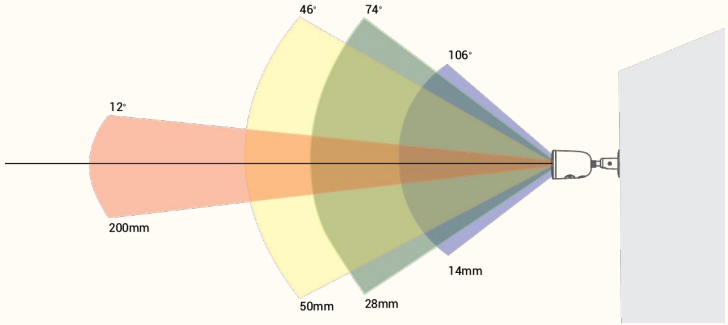
The human eye can only distinguish about 200 to 256 different levels of brightness under normal viewing conditions.



Image from Prof. Srinivasa Narasimhan’s slide

PART I: Image Formation & Acquisition

Field of View (FOV)











(H/V): 70°/55°



(H/V): 80°/65°



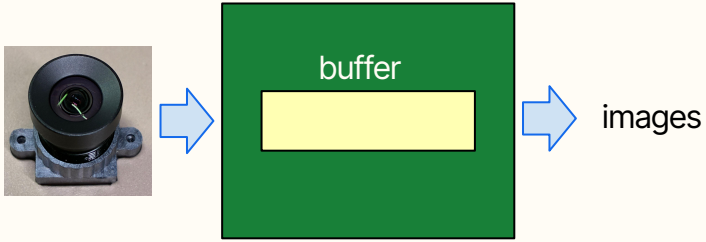
(H/V): 110°/94.5°

PART I: Image Formation & Acquisition

Frames Per Second (FPS)

Frequency at which consecutive images (frames) are displayed

Arducam OV9281 camera runs
- 45 FPS for 1280 x 800 images
- 140 FPS for 640 x 400 images



Exposure Rate

The amount of light used to expose a photograph



Higher exposure @ 1 FPS



Normal exposure @ 45 FPS

PART I: Image Formation & Acquisition

Let's look at the code: Image Pointwise Processing

- Reading and viewing the image
- Understanding how images are interpreted in computers
- Image intensity histogram
- Increasing image intensity contrast
- Image thresholding

https://github.com/Brown-LEMS/BREADTH_Scholars_Curriculum_Computer_Vision



PART II: Image Convolution

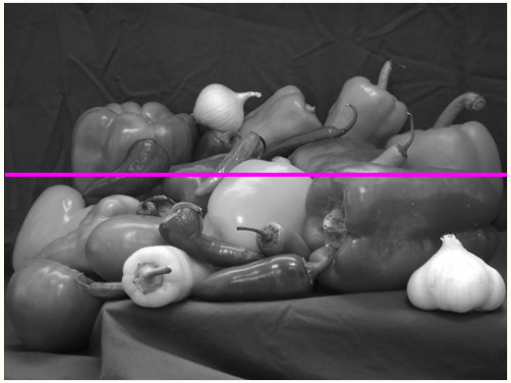
Image signals are often corrupted with *noise*

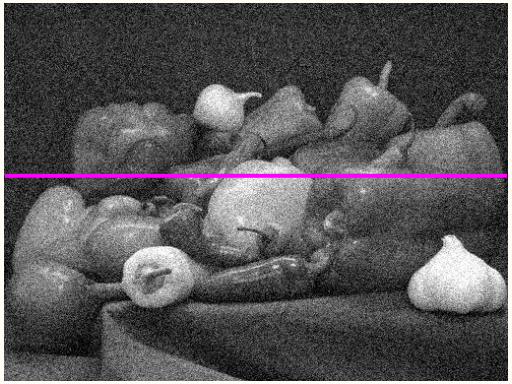
- Irregularity: lacks continuity and appears as rapid, random fluctuations without correlation across neighboring points, exhibiting a wide spectrum of *frequencies*

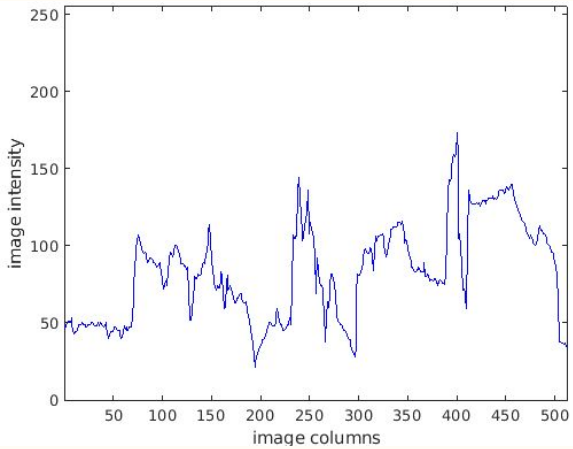


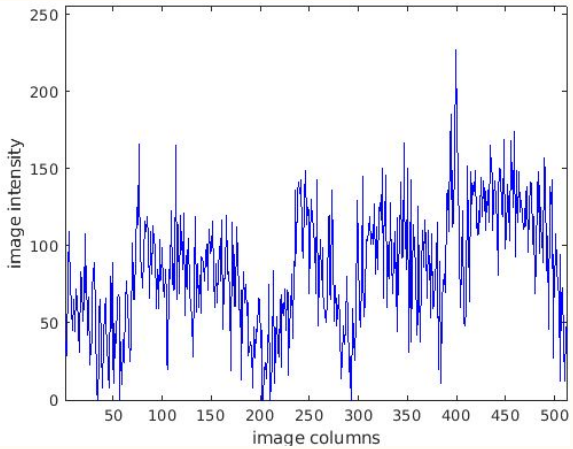


PART II: Image Convolution

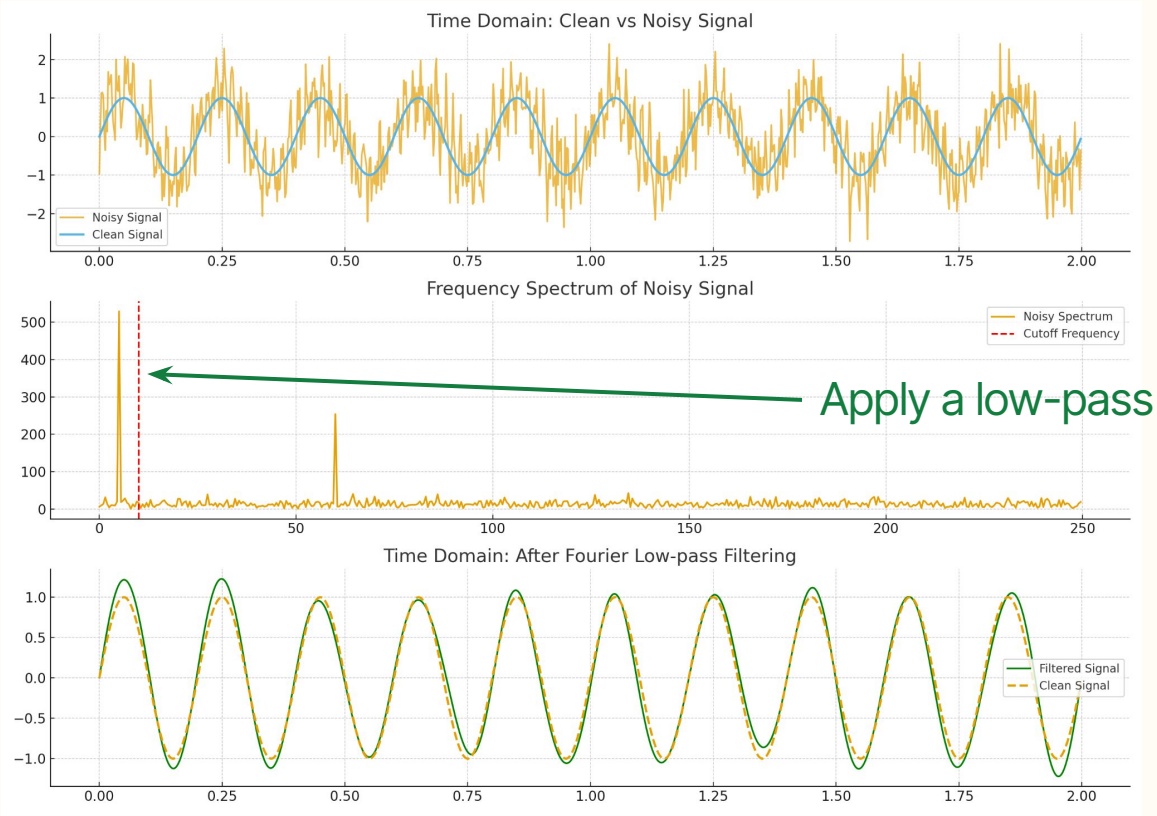




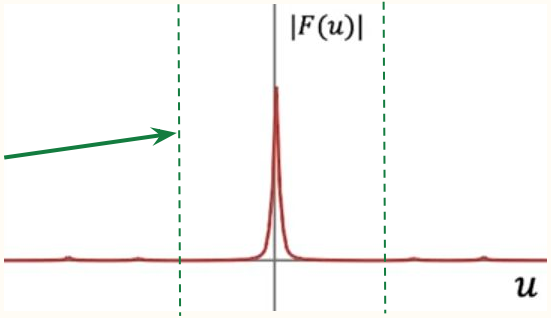




PART II: Image Convolution



Apply a low-pass filter



PART II: Image Convolution

The change between time and frequency domains is captured by Fourier transform

$$F(\omega) = \int_{-\infty}^{\infty} f(x) e^{-i\omega x}\, dx$$

signal in frequency domain — signal in time domain

Let's say we have a low-pass filter *H(w)* so that

$$G(\omega) = F(\omega)H(\omega)$$

which in time domain

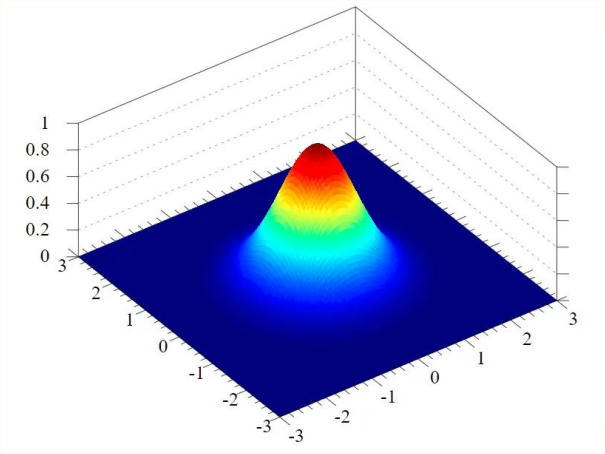
$$g(t) = f(t) * h(t)$$

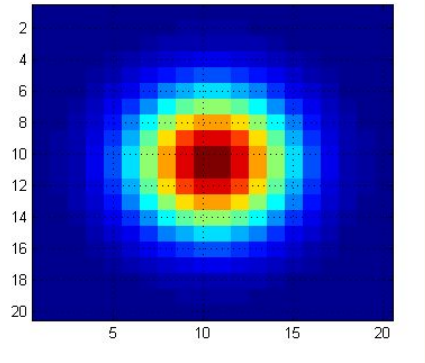
convolution!

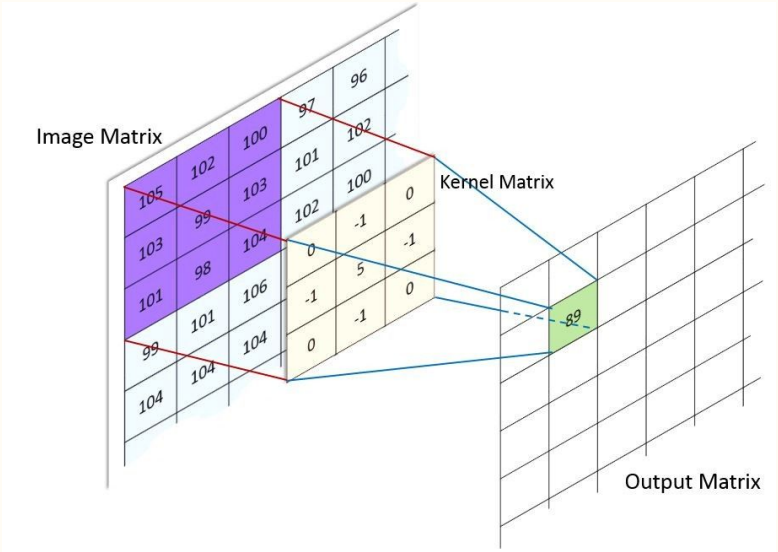
A good low-pass filter is a Gaussian function

PART II: Image Convolution

On an image, we need a 2D discrete Gaussian kernel







- Other types of kernels extract features on an image
  - Convolutional Neural Network (CNN) learns the kernel to extract image features for various applications

PART II: Image Convolution

Let's look at the code: Image Noise Removal by Convolution

- Removing noise from the image
- Evaluate the effectiveness of Gaussian convolved images with an edge detector
- Explore convolution through separable 1D Gaussian kernels

https://github.com/Brown-LEMS/BREADTH_Scholars_Curriculum_Computer_Vision

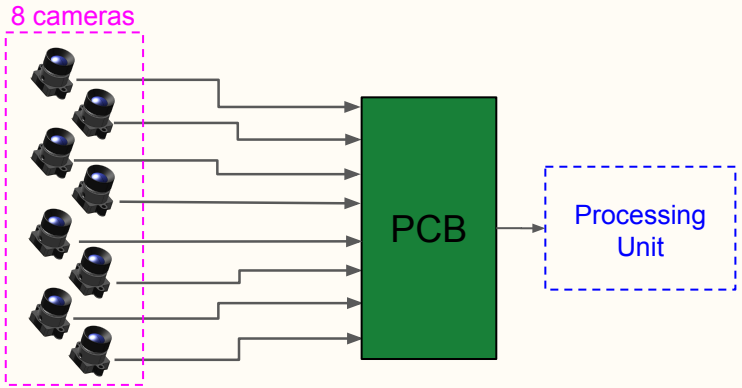


PART III: Multi-Camera Image Synchronization

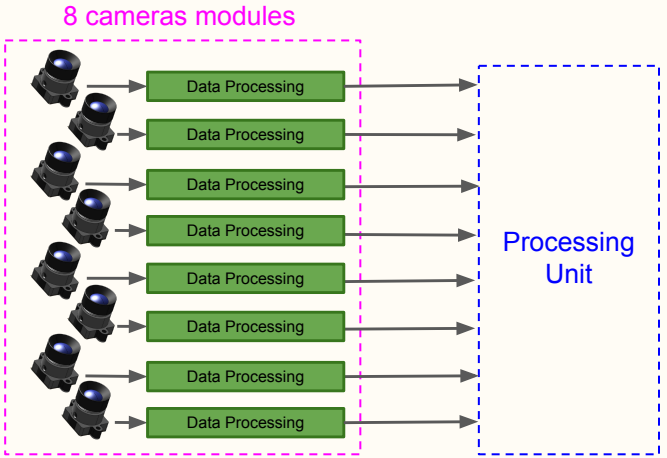
Synchronizing images is critical!

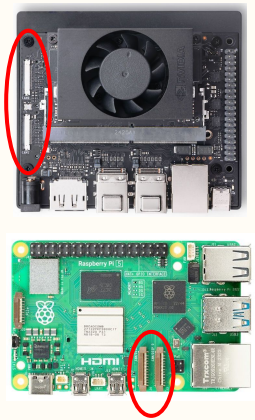
- Multiple evidence from multiple cameras have to support the “same” event

Hardware-level Synchronization



Software-level Synchronization



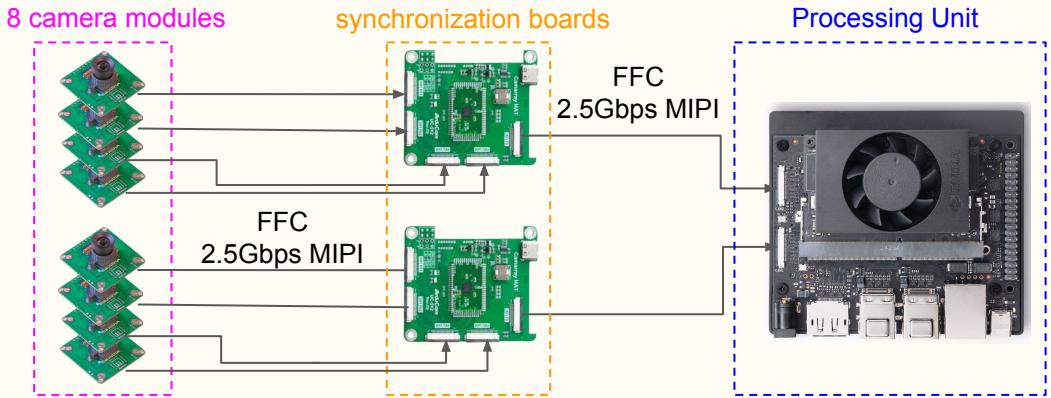


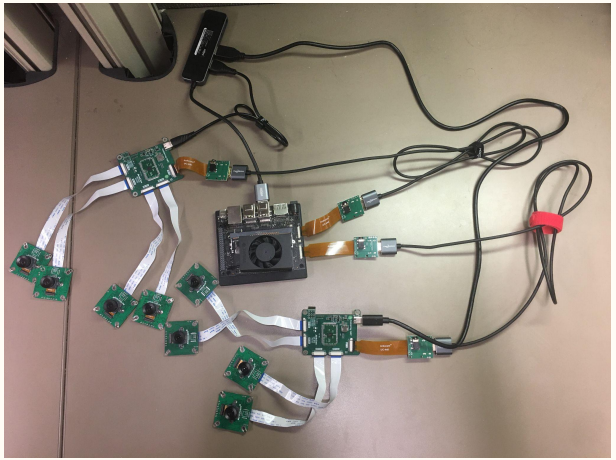
Major problem: interface limit + bandwidth limit

PART III: Multi-Camera Image Synchronization

We have somewhat between the hardware and software level synchronization

- Hardware-level synchronization for a set of four images

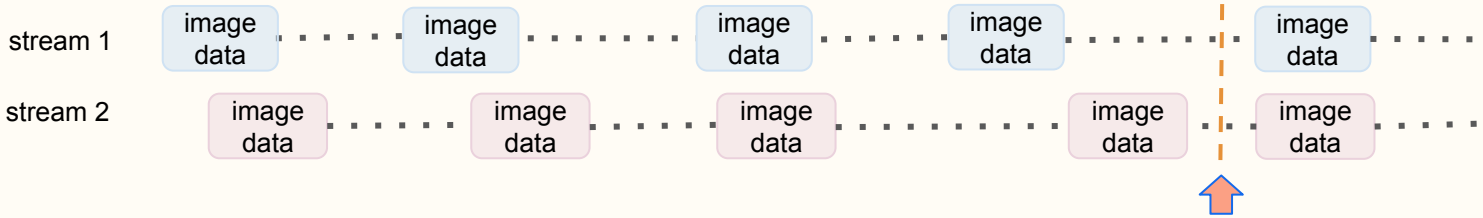




PART III: Multi-Camera Image Synchronization

We have somewhat between the hardware and software level synchronization

- Software-level synchronization for two (stacked) images



network clock alignment + coordinated acquisition-thread triggering + camera phase correction

PART III: Multi-Camera Image Synchronization

We have somewhat between the hardware and software level synchronization

- The time difference between the two stacked images is less than 0.1-0.25 ms

